

AI-Generated vs Real Image Classification

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Abstract—The growing realism of generative models has made AI-generated images increasingly hard to distinguish from real photographs, motivating reliable automated detection. This work studies the binary classification of real versus AI-generated images at low spatial resolution using the CIFAKE dataset, which pairs CIFAR-10 photographs with Stable Diffusion 1.4 generations. Two convolutional architectures are compared: a compact custom CNN (0.36 M parameters) and a ResNet-18 trained from scratch (11.2 M parameters). Both exceed 96% test accuracy on CIFAKE (AUC 0.994 and 0.995), with the deeper network leading by only 0.5 percentage points. Crucially, when evaluated on the Defactify dataset spanning five unseen generators, accuracy collapses to 0.43–0.67 and falls to chance on DALL-E 3 and Midjourney v6, with precision degrading while recall stays high. This shows that the detectors internalise a generator-specific decoder artefact rather than a general notion of synthetic imagery, and demonstrates that high single-source benchmark accuracy substantially overstates real-world detection utility.

Index Terms—Convolutional Neural Network, Residual Neural Network, Artificial Intelligence, Image Generation

I. INTRODUCTION

Recent advances in generative models have made it possible to synthesise photorealistic images that are increasingly difficult to distinguish from real photographs. As these tools become widely accessible, the volume of AI-generated imagery circulating online has grown rapidly, raising concerns across journalism, legal evidence, scientific documentation, and public discourse.

The difficulty of this task for people was documented by Roca et al. [1], who ran a large controlled experiment measuring how well humans detect AI-generated images. They found that human accuracy is limited and varies substantially with image content and generator quality. The participants were overall only 62% accurate, which is only slightly better than flipping a coin. This result demonstrates that human judgement alone is not a dependable safeguard against synthetic media, and it establishes a need for more accurate detection mechanisms.

These points are the motivation for this project, which investigates and compares machine learning approaches for classifying real against AI-generated images.

II. STATE OF THE ART REVIEW

The task of distinguishing AI-generated from real images has been approached through multiple ideas, which this section covers.

Bird and Lotfi [2], who introduced the CIFAKE dataset, trained a Convolutional Neural Network directly on raw pixel

data to classify real CIFAR-10 images against Stable Diffusion 1.4 generations, reaching high accuracy on the binary task. They applied Grad-CAM to interpret the classifier, finding that predictions on real images drew on the whole image, whereas predictions on AI-generated images concentrated on localised, artefact-prone regions in both the subject and the background. This established that local convolutional features alone capture much of the real/fake signal.

A different approach was taken by Aziz et al. [3], who compared low-level image properties between camera photographs and outputs from several generative models, including Stable Diffusion. They reported that AI-generated images tend to exhibit higher hue, saturation, and brightness than real camera images, indicating that simple colour statistics carry discriminative information about an image's origin.

A paper by Sha et al. [4] showed that images produced by text-to-image diffusion models contain detectable digital fingerprints, which are statistical traces left by the generation process. They demonstrated that these traces can be learned to both detect fakes and attribute them to the specific generator that produced them. Corvi et al. [5] supports this concept, since their analysis of the Fourier spectrum of generated images found that diffusion-model fingerprints are directly visible as grid-like artefacts at specific frequencies. These spectral peaks are largely absent from real photographs, meaning that the generation process leaves discriminative traces in the frequency domain. This could be exploited either explicitly, by defining new features for this data, or implicitly by pixel-domain CNNs, relying on their learning of the structure.

Besides CNNs, alternative models have also been used. He et al. [6] introduced residual networks (ResNet), whose skip connections enable training of very deep CNNs. ImageNet-pretrained ResNets are widely fine-tuned for image-forensics and real/fake classification. Dosovitskiy et al. [7] proposed the Vision Transformer (ViT), which replaces convolution with patch-based self-attention and a global receptive field, enabling modelling of long-range correlations and global consistency violations characteristic of synthetic imagery.

III. DATA DESCRIPTION, VISUALIZATION AND STATISTICAL ANALYSIS

A. Problem

This work studies the discrimination of real photographs from AI-generated images on low spatial resolution. There are several known generators, but during this report the focus is on Stable Diffusion 1.4 [8]. The central question is whether such low-resolution images retain a signal sufficient to discriminate the two classes reliably.

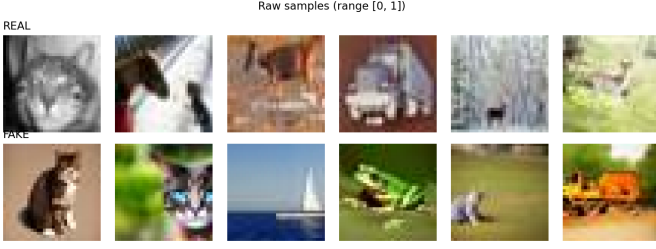


Fig. 1. Random training samples. Top row: REAL (CIFAR-10 photographs); bottom row: FAKE (Stable Diffusion 1.4 generations).

B. Source and class balance

The selected dataset is CIFAKE [2], which pairs the CIFAR-10 image database (real images) with synthetic counterparts produced by Stable Diffusion 1.4. The released split provides 100,000 training images (50,000 per class) and 20,000 test images (10,000 per class). Classes are perfectly balanced, so no resampling or class weighting is required. A fixed-seed 20% slice of the training partition is held out for validation, leaving 80,000 images for actual training and reserving the 20,000-image test partition for the final evaluation only.

C. Visualization

Figure 1 shows a random subset of unprocessed images from each class. At 32×32 resolution, both classes are visually indistinguishable to the naked eye, which hints at the difficulty of the task.

D. Statistical analysis

To check whether the two classes differ beyond what the eye can see, the average 2D Fourier log-magnitude spectrum of 5,000 samples per class was computed (Figure 2). The spectrum measures how much energy each spatial frequency carries: the centre captures broad structure, the corners capture fine texture.

The two means agree at low frequencies but FAKE carries noticeably more high-frequency energy than REAL, a known fingerprint of latent diffusion’s image decoder [5]. The signed-difference panel on the right of Figure 2 makes this asymmetry visible: red cells mark spatial frequencies where the FAKE class exceeds REAL, blue cells mark the reverse, and colour saturation encodes the magnitude of the gap. The dominant red “wing” across mid-to-high frequencies, together with the small blue blob at the centre, shows that REAL only slightly dominates at the lowest spatial frequencies while FAKE dominates almost everywhere else. The signal is present *before any model is trained*, confirming that CIFAKE retains a detectable cue despite the low resolution.

E. Data quality and limitations

The released archive is pre-cleaned: every image loads successfully, all are uniform 32×32 RGB, and no duplicates or zero-byte files were detected. No additional cleaning was applied. Two structural limitations frame the rest of the study: (i) the FAKE half originates from a single generator (Stable

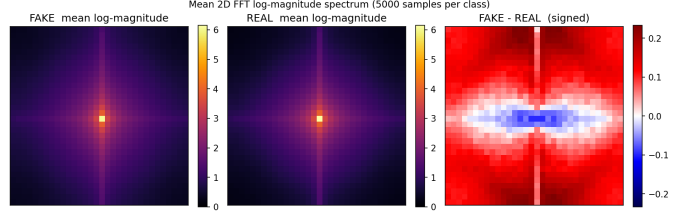


Fig. 2. Mean 2D FFT log-magnitude spectrum per class. Left: FAKE. Centre: REAL. Right: signed difference.

Diffusion 1.4), so generalisation is not guaranteed; (ii) the low spatial resolution suppresses high-frequency cues that higher-resolution detectors typically exploit.

IV. DATA PREPROCESSING

Preprocessing is minimal. Since the dataset already has perfect 50-50 class split, no resampling is required. No image resizing is needed either because inputs are kept at the native 32×32 resolution and the chosen models natively accept 32×32 images.

A. Augmentation

We applied a small number of augmentation techniques due to the findings present in Section III. As we wanted to avoid destroying the frequency-domain artefacts encountered, most strategies such as blurring or colour jitters were excluded from the preprocessing pipeline. The only modifications we apply are random horizontal flipping and a random rotation of $\pm 10^\circ$ because their effect on the Fourier spectrum is minimal.

To stimulate data variability not only at initialisation but throughout training, these transforms are reapplied to each image in real time at every epoch. This exposes the CNN¹ to fresh positional and orientation variants each pass, encouraging invariance to such changes and reducing overfitting.

B. Normalization

Z-score standardisation is applied for each color channel. The values for the mean and standard deviation were computed once over the full training partition and cached:

$$\mu = (0.472, 0.463, 0.418), \quad \sigma = (0.238, 0.237, 0.266).$$

The same μ and σ are reused for all images during training, as well as at validation and test time, and are persisted alongside each model checkpoint to guarantee that normalisation is identical between training and evaluation.

To assess whether normalisation actually contributes to the final score, each architecture is also trained without it. The comparison is reported in Section VI.

¹ResNet included, as a specific type of CNN.

V. DESCRIPTION OF THE APPLIED MACHINE LEARNING ALGORITHM(S)

A. Problem Statement

The task is *binary supervised classification*. Each input is a single RGB image, i.e. a vector of $3 \times 32 \times 32 = 3072$ pixel intensities in $[0, 1]$. The model produces two real-valued logits, converted to a probability vector by softmax. The predicted label is taken as the $\arg \max$ of this vector, equivalent to thresholding $P(y = \text{REAL} \mid x)$ at 0.5.

B. Selected models

From the state-of-the-art review of Section II, the following two convolutional architectures were selected for this problem.

1) *Convolutional Neural Network (CNN)*: A custom baseline uses three convolutional blocks of the form [Conv 3×3 —BN—ReLU—MaxPool2], with channel counts growing $3 \rightarrow 32 \rightarrow 64 \rightarrow 128$, followed by a two-layer fully-connected head with dropout 0.3. The compact depth is motivated by the very low spatial resolution of the input. Total parameter count: 0.36 M.

2) *ResNet-18*: A deeper alternative is the 18-layer residual network of He *et al.* [6], the smallest standard variant in the ResNet family. Following the CIFAR adaptation in the same paper, the ImageNet stem was replaced by a single 3×3 convolution. The fully connected layer is reduced to two logits, and all weights are randomly initialised, ensuring that the model is trained from scratch. Total parameter count: 11.2 M.

C. Training procedure

Each model is trained with Adam optimiser, cross-entropy loss, a batch size of 256, weight decay (CNN: 10^{-4} , ResNet-18: 10^{-6}) and a fixed random seed of 42 for reproducibility. After every epoch, training and validation loss and accuracy are recorded. The checkpoint with the highest validation accuracy is kept.

1) *Early stopping*: Training is aborted when validation accuracy fails to improve for three consecutive epochs (patience = 3). Combined with the best-checkpoint policy above, this guards against overfitting without committing to a fixed epoch count.

D. Hyperparameter search

Learning rate (η) was considered the most influential hyperparameter since it sets the base step that Adam scales per-parameter. Each model was tested with learning rates log-spaced over four orders of magnitude, $\{10^{-5}, 3 \cdot 10^{-5}, 10^{-4}, 3 \cdot 10^{-4}, 10^{-3}, 3 \cdot 10^{-3}, 10^{-2}\}$. For every value, the model was trained for ten epochs on a 20% deterministic subsample of the training partition, and the best validation accuracy was recorded. The reduced budget keeps the sweep cheap while separating clearly good rates from clearly bad ones.

Figure 3 reports the result. Both architectures peak in the same region: the CNN at $\eta = 10^{-3}$ (93.7%) and ResNet-18 at $\eta = 3 \cdot 10^{-4}$ (94.1%).

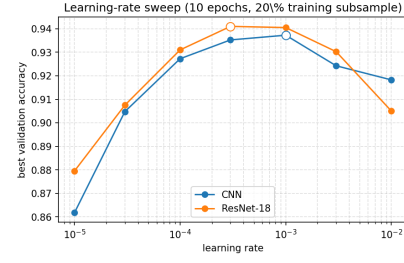


Fig. 3. Learning-rate sweep for both architectures. Best validation accuracy reached after ten epochs on a 20% training subsample. White markers indicate the chosen operating point per model.

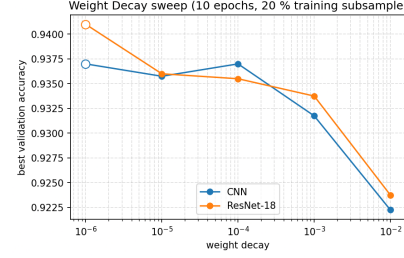


Fig. 4. Weight-decay sweep for both architectures. Same protocol as Figure 3. White markers indicate the selected operating point per model.

The same protocol was applied to weight decay over $\{10^{-6}, 10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}\}$ (Figure 4). The CNN sits on a plateau between 10^{-6} and 10^{-4} , while ResNet-18 peaks at 10^{-6} . The per-model optima (10^{-4} for the CNN, 10^{-6} for ResNet-18) are adopted for the final runs in Section VI.

VI. PRESENTATION AND DISCUSSION OF RESULTS

A. Test performance

Table I reports the held-out test metrics of both architectures, trained on the full 80,000-image training pool with the learning rates selected in Section V-D and evaluated on the untouched 20,000-image test partition.

Both models clear 0.96 test accuracy. ResNet-18 wins on four out of five metrics by margins of 0.1 to 1.6 percentage points. The CNN keeps the lead on recall by 0.6 points. The result is consistent with the intuition that depth helps, but the gap is small, which is itself an informative finding: a 0.36 M-parameter network with three convolution blocks is only 0.5 points behind a 11.2 M-parameter residual network on the same data. This suggests that the dataset does not require deep representational capacity.

B. Training dynamics, overfitting and underfitting

Figures 5 and 6 show the per-epoch training and validation curves. The CNN trains for 24 epochs and ends with a training loss of 0.09 and a validation loss of 0.11, with most epochs keeping the gap below 0.02 — a controlled-fitting regime with neither overfitting nor underfitting. ResNet-18 trains for 18 epochs, reaches a training loss of 0.05 while validation flattens around 0.10, leaving a ≈ 0.05 gap. Validation accuracy then plateaus and early stopping triggers, retaining the best checkpoint from epoch 15 and validating the criterion introduced in Section V-C1.

TABLE I
TEST-SET PERFORMANCE ON THE 20,000-IMAGE CIFAKE TEST
PARTITION. BEST PER COLUMN IN BOLD. TIME MEASURED ON A SINGLE
RTX 5060.

Model	Acc.	Prec.	Rec.	F1	AUC	Time (s)
CNN (0.36 M)	0.962	0.952	0.972	0.962	0.994	94
ResNet-18 (11.2 M)	0.967	0.968	0.966	0.967	0.995	377

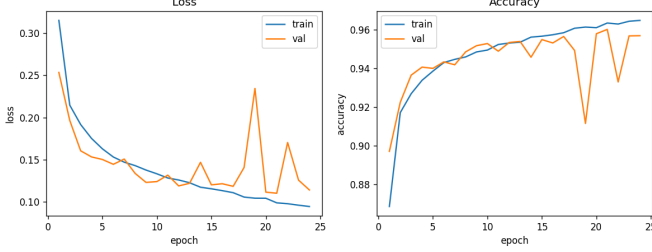


Fig. 5. CNN – per-epoch training and validation loss and accuracy.

C. Confusion matrices and per-class behaviour

Figure 7 shows the 2×2 confusion matrices for both models. The two classifiers display the same asymmetric error profile. False positives (FAKE predicted as REAL) are slightly more frequent than false negatives. The default 0.5 threshold is not necessarily optimal. The ROC analysis below explores the threshold trade-off explicitly.

D. ROC curve and threshold trade-off

Figure 8 overlays the ROC curve of each model. Each point on the curve corresponds to a different value of the decision threshold applied to $P(y = \text{REAL} \mid x)$. Both curves stay very close to the top-left corner, which means that across essentially every threshold choice both models keep a high true-positive rate at a low false-positive rate, meaning they separate the two classes cleanly regardless of where the threshold is placed. The AUC values, 0.994 for the CNN and 0.995 for ResNet-18, summarise this in a single number and confirm that the gap observed in Table I is genuine ranking quality rather than an artefact of the default 0.5 cut-off. The threshold itself is also a useful lever at deployment time. Sliding the operating point leftwards along the curve trades a small amount of recall for a much lower false-positive rate, which can be exploited when the cost of a false alarm is higher than that of a missed detection.

E. Effect of normalisation

Table II reports the test metrics of both architectures with and without per-channel Z-score normalisation. The effect is asymmetric: removing it costs the CNN 3.5 percentage points of accuracy and nearly 6 of precision, while ResNet-18 loses only 0.1 points and is indistinguishable on AUC. Because both networks use batch normalisation internally, the deeper residual network has enough representational slack to absorb a poorly scaled input, whereas the shallow CNN does not. Normalisation thus behaves less as a generic regulariser than as a compensation for limited model capacity, and is kept enabled in every reported configuration.

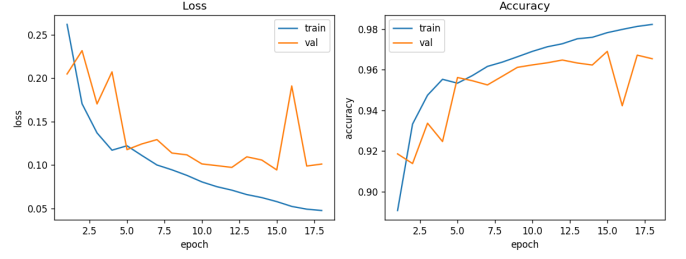


Fig. 6. ResNet-18: per-epoch training and validation loss and accuracy. Training loss falls close to zero while validation flattens, indicating mild overfitting and triggering early stopping.

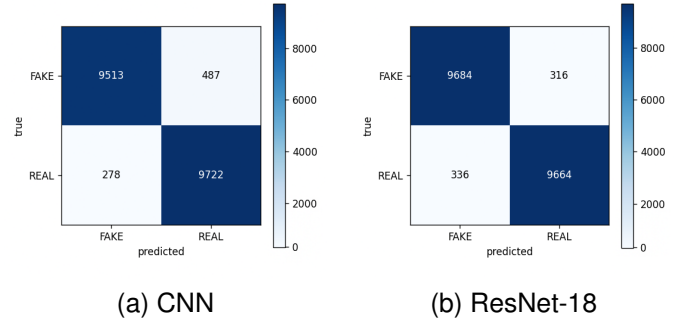


Fig. 7. Confusion matrices on the 20,000-image test partition. Rows are true labels, columns are predicted labels.

F. Spatial attention via Grad-CAM

Grad-CAM [9] was applied to the last convolutional block of both models on the same eight test inputs to make the attention patterns directly comparable (Figure 9). The CNN concentrates on small, focal patches, while ResNet-18 lights up a broader region around the subject. Bird and Lotfi [2] report that their custom CNN attends to the whole image on REAL samples and to localised artefacts on FAKE ones. Our CNN row matches this description, but the ResNet-18 row spreads attention more uniformly across both classes, suggesting that the deeper network does not rely on a localised cue to reach its decision.

G. Cross-generator generalisation

To probe whether the detector transfers beyond Stable Diffusion 1.4, both models were evaluated on the Defactify [10] test split, which pairs MS-COCO photographs with synthetic images from five generators (Stable Diffusion 2.1, SDXL, Stable Diffusion 3, DALL-E 3 and Midjourney v6). For each generator, 500 synthetic plus 500 real images were sampled, resized to 32×32 , and normalised with the training statistics. Table III reports the metrics.

Accuracy drops from above 0.96 on CIFAKE to 0.43–0.67 on Defactify, degrading with the distance from the training source: SD 2.1 and SD 3 share the latent-diffusion family and remain partially detectable (best AUCs 0.69 and 0.80), while DALL-E 3 and Midjourney v6 sit close to or below chance. Recall stays high (0.78–0.93) and precision collapses (0.46–0.61), so the classifier defaults to REAL on out-of-distribution inputs. Combined with the frequency-domain sig-

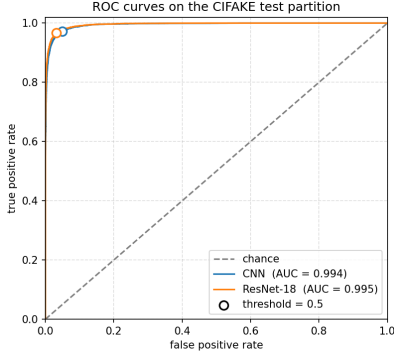


Fig. 8. ROC curves on the CIFAKE test partition for both models. The dashed grey line is the chance reference. AUC values are reported in the legend.

TABLE II
EFFECT OF DATASET-WIDE Z-SCORE NORMALISATION ON TEST-SET METRICS. BEST PER MODEL IN BOLD.

Model (norm)	Acc.	Prec.	Rec.	F1	AUC
CNN, with	0.962	0.952	0.972	0.962	0.994
CNN, without	0.926	0.894	0.966	0.929	0.983
ResNet-18, with	0.967	0.968	0.966	0.967	0.995
ResNet-18, without	0.966	0.970	0.962	0.966	0.995

nature of Section III, this points to the network having internalised an SD 1.4 decoder artefact: CIFAKE-trained detectors do not transfer to arbitrary generators.

The DALL-E 3 and Midjourney v6 AUCs sit too close to the 0.5 chance line to justify inverting the classifier’s predictions. On these generators, manual review or even flipping a coin might give better results.

VII. CONCLUSIONS

The cross-generator evaluation is the most consequential result of the study. A detector scoring above 0.96 on CIFAKE collapsed to 0.43–0.67 accuracy on unseen generators, falling to chance on DALL-E 3 and Midjourney v6. The systematic precision collapse with sustained high recall shows the models tend to predict REAL on out-of-distribution inputs, so it seems they have memorised the Stable Diffusion 1.4 decoder artefact rather than learning a generator-agnostic notion of “synthetic”. This reframes the contribution: high benchmark accuracy on a single-source dataset overstates real-world utility, and CIFAKE-trained detectors must not be deployed against arbitrary generators without retraining.

A. Future work

Three directions stand out for future work. The first targets the resolution bottleneck. Training and evaluating at the generators’ native resolution, rather than 32×32 , would preserve the high-frequency fingerprint that is hidden by the downsampling step.

The second addresses the data side. Training on a multi-generator dataset would test whether generator-agnostic detection is achievable through diversity alone. The Defactify dataset [10] already provides data from five generators so it may be used as a starting point.

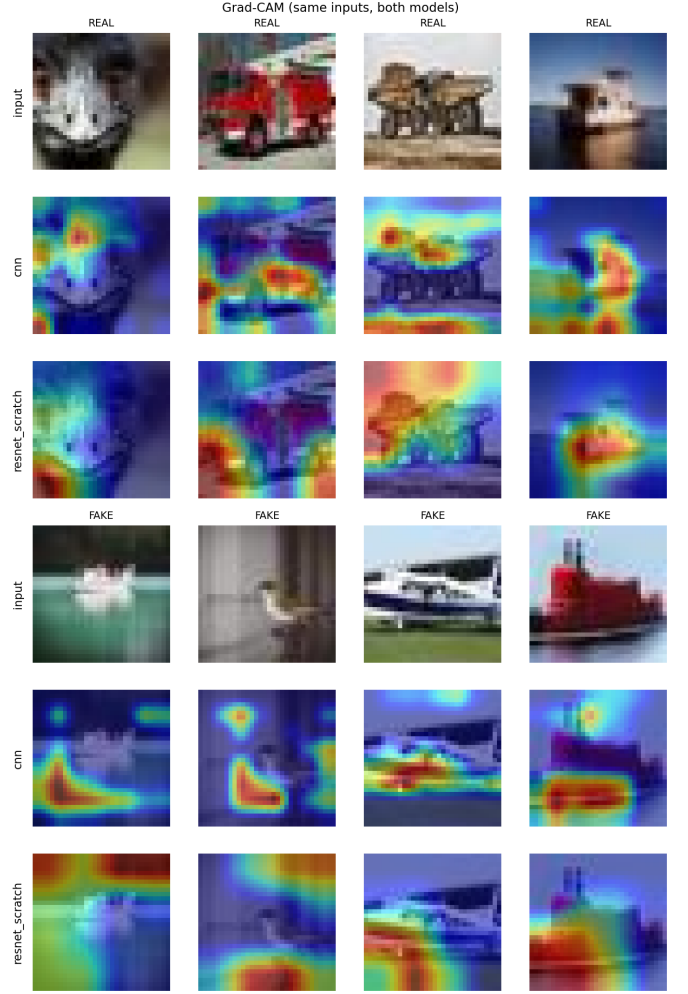


Fig. 9. Grad-CAM heat-maps for both models on the same test inputs. Top band: REAL; bottom band: FAKE.

TABLE III
CROSS-GENERATOR EVALUATION ON THE DEFACIFY TEST SPLIT. EACH ROW USES 500 REAL PLUS 500 FAKE IMAGES OF THE INDICATED GENERATOR, RESIZED TO 32×32 TO MATCH THE MODEL INPUT. BEST PER (GENERATOR, METRIC) IN BOLD.

Generator	Model	Acc	Prec	Rec	F1	AUC
SD 2.1	CNN	0.560	0.535	0.912	0.675	0.686
	ResNet-18	0.566	0.543	0.828	0.656	0.634
SDXL	CNN	0.537	0.521	0.932	0.668	0.626
	ResNet-18	0.566	0.545	0.806	0.650	0.630
SD 3	CNN	0.671	0.613	0.928	0.738	0.799
	ResNet-18	0.614	0.581	0.822	0.680	0.675
DALL-E 3	CNN	0.428	0.460	0.828	0.591	0.457
	ResNet-18	0.464	0.478	0.792	0.596	0.487
Midjourney v6	CNN	0.449	0.472	0.846	0.606	0.446
	ResNet-18	0.467	0.480	0.784	0.595	0.538

The third concerns the choice of architecture. The Vision Transformer was excluded because at 32×32 resolution, 16×16 patches yield only four tokens, making self-attention trivial and negating the architecture’s advantage over CNNs. At higher native resolution (step 1), this limitation no longer applies and ViT may produce better results on the task.

VIII. ETHICAL CONSIDERATIONS

A high-accuracy detector for AI-generated images is dual-use. The same tool that can defend against disinformation can be repurposed to discredit authentic content, which is a particular risk for minorities, political dissidents, or independent journalists who may not have the resources to contest an automated flag. The asymmetric error profile reported in Section VI amplifies this risk: a deployed instance of our CNN would mistake roughly 5% of real photographs for AI generations, an unacceptable rate without human review. The collapse on unseen generators in Table III compounds the problem, because deployment outside the Stable Diffusion 1.4 family could produce confident but systematically wrong predictions. We therefore consider that responsible deployment of a detector of this kind should report per-class precision and recall on representative data, calibrate the decision threshold per use case rather than relying on the default 0.5 cut-off, gate automated actions behind a high-confidence floor, and route low-confidence cases to human review. The underlying training data should also be documented so users can assess whether the generator they care about is even in distribution.

IX. CRITICAL REFLECTION

A. Many generators exist

In practical terms, our results are only useful against the specific generator the model was trained on, which essentially invalidates deployment in any realistic scenario where the generator behind a given image is not known in advance or when the generator is not Stable Diffusion 1.4. Looking back at the project, the decision to evaluate on a second, multi-generator dataset was what made this limitation visible. Had we relied solely on the CIFAKE test partition, we would have reported a strong classifier without ever noticing how brittle it actually is. The main lesson is that benchmark numbers on a single source can be deeply misleading, and that any future iteration of this work should treat cross-generator evaluation as a primary metric rather than an afterthought.

B. New generators will keep being created

The results made clear why this problem is a moving target. Each generator has its own signature, so a detector built around one of them might be mostly learning that one signature and not a general idea of what synthetic images look like. This connects the technical side to the bigger picture, because detection tools are always chasing generators that keep changing through new versions and algorithms. It shows that this is not a problem that gets solved once, but one that has to be maintained as the technology behind image generation keeps improving.

C. Input size requirement

Another aspect is that images are required to be 32×32 . This is not just a methodology detail but a fundamental issue for any practical deployment of the low-resolution detector. Real images come at arbitrary resolutions and the resize process might hide the cue the models rely on. A possible solution

would be to apply the models on cropped patches of the image at the original resolution, so the high-frequency fingerprint is preserved at the cost of more inference calls per image.

X. AI TOOLS

This work made use of several AI assistants, summarised in Table IV. Each was treated as a tool whose outputs were inspected and validated by the authors before inclusion. All numerical results reported in this document were produced by the authors' code on the authors' hardware. No experimental finding, table value, or figure was generated by an AI assistant.

TABLE IV
AI ASSISTANTS USED DURING THE PROJECT.

Tool	Purpose	Rationale
Claude	Code development support.	Strong and fast code understanding used to easily spot logic flaws.
Claude	Writing review and phrasing suggestions.	Fast and detailed spelling check and cohesion validation.
Gemini	Reference discovery for the SOTA review.	Google Search grounding with inline source links.

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